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| **PI Name:** | Smilenov, Lubomir Ph.D. | | |
| **Project Title:** | The Effect of High LET Radiation on Differentiation and Tumorigenesis in the Human Hematopoietic System: Modeling in vitro and in vivo for Risk Assessment | | |
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| **Task Description:** | The goal of the study is to contribute to the high-LET (linear energy transfer) radiation carcinogenesis risk estimation. This goal will be achieved using in vivo data acquired from an irradiated human hematopoietic system reconstituted in immunodeficient mice. The study is based on a stepwise enrichment strategy the first stage of which will be to irradiate mice engrafted with human hematopoietic system followed 5-6 months later by analysis of human stem and progenitor cells from mouse bone marrow for chromosomal damage and developmental abnormalities. The goal in this initial stage will be to identify human pre-leukemic stem cells and progenitor cells which exhibit altered karyotypes and abnormal differentiation patterns. The second stage involves engraftment of human hematopoietic stem cells with known abnormalities in secondary mouse hosts for further tumor developmental studies. This secondary host transplantation strategy, which is commonly used in human tumor engraftment studies, will allow the enrichment of the engrafted mouse cohorts for potentially tumorigenic carriers and permit elimination of the mice with normal karyotypes which usually don't develop tumors. We expect the planned studies to show, for the first time, the effects of high LET radiation on a live human hematopoietic system providing information on 1) radiation induced carcinogenesis, 2) induction of in vivo chromosomal aberrations of human hematopoietic stem and progenitor cells, and 3) in vivo ablation of human hematopoiesis. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The effects of spaceflight on human physiology are complex and represent a sum of many factors, the main two of which are microgravity and radiation. The effect of exposure to the space environment is illustrated in studies that show altered immune response of T lymphocytes following spaceflight dating back to Apollo and Skylab missions. Although this is reversed once the astronauts are back to the Earth, these effects are persistent in space and were more recently documented in astronauts during their six month mission to the International Space Station. While difficult to estimate which one of the stress factors is dominant in low Earth orbit flights, a flight to Mars for example will lead to radiation exposure that is above NASA's current standards and its effect has to be studied in advance. The human hematopoietic system is one of the most radiosensitive tissues in the human body given that it is comprised of a large number of continuously and rapidly proliferating cells. It is well known that an insult to the hematopoietic system will generate effects that are not limited in scope to hematologic endpoints. There is an interrelationship between the correct function of the hematopoietic system and other organ systems. All of this knowledge taken together has sparked much scientific effort on studies of the effects of radiation on human hematopoietic system in the last decades using in vitro and in vivo studies in model organisms. A noticeable gap in these studies (and for obvious reasons), is the lack of in vivo studies of radiation effects on the human hematopoietic system. The recent development of immunodeficient mouse models allows for first time such studies. These mouse models allow the engraftment of human hematopoietic stem cells in the mouse bone marrow, leading to reconstitution of functioning human hematopoietic system in the mouse bone marrow, spleen, thymus, and blood. Importantly, the side effect for the mice is minimal. This permits long term human cell engraftment and gives excellent opportunity for radiation studies. To the best of our knowledge, the proposed studies will be the first that will evaluate the effects of high- and low-LET radiation in an engrafted human immune system. We expect that the planned studies will show, for the first time, in vivo data detailing the early effects from high LET radiation. Our work will also permit greater understanding of the cancer causing biology that takes place in a complete human hematopoietic system after high LET radiation exposure. We believe that the acquired knowledge will reduce the risk of astronauts in long term missions. |
| | FINAL REPORTING--SEPTEMBER 2017<br>Final task book report<br><br>Our main hypothesis is that studies of the effects of high-LET radiation on a human hematopoietic system, reconstituted in immunodeficient mice, will provide meaningful information on not only post-radiation depletion and repopulation kinetics but also tumor formation. Such information can then be applied to space radiation risk assessment. The importance of this work is clear given the lack of in vivo data on the effects of high LET radiation on human organ and tissue systems. The study has two key objectives: The first is to characterize the early and late response of a human immune system reconstituted in mice to high LET radiation and gamma-rays. The goal here is to build a detailed picture of the series of changes that the reconstituted human hematopoietic system undergoes after a high LET radiation exposure. The second objective is to characterize the human tumors that may arise in the irradiated and non-irradiated mice. To accomplish these objectives we produced and irradiated mice engrafted with a human hematopoietic system. The mice were exposed to either HZE (high energy) ions or X-rays and subsequently monitored along with non-irradiated control mice. The early effects of the radiation on the human cells are assessed by periodic peripheral blood tests. The obtained differential counts are compared with differentials acquired before the irradiation. Monitoring the peripheral blood cell counts for extended periods of time may show the depletion and recovery kinetics of peripheral blood cells which will be a measure of the degree and duration of the radiation induced damage and also the ability of the human hematopoietic system to recover after radiation induced insults. In addition, the mice are monitored for tumor development. The tumor characterization was done by a team of pathologists from the Columbia University Department of Pathology and Cell Biology using established clinical laboratory methods for human tumor characterization with corrections based on the specificity of the mouse model. This project directly addresses high-priority research topics emphasized by the Space Radiation Program Element for FY2013 and in in particular the Risk of Radiation Carcinogenesis from Space Radiation and Risk of Degenerative Tissue or other Health Effects from Space Radiation.<br><br>Assessment of the early and late effects of high LET 28Si irradiation and X-ray irradiation on the depletion and recovery kinetics of human peripheral blood cells in humanized mice: Clinical and physiologic assessment of any radiation exposure entails a survey of circulating lymphocyte counts. This is no different with high LET radiation and so we assess the response of engrafted human cells by examining these blood cell counts. Two very important endpoints can be measured by determining the blood cell counts at different periods after the irradiation. These are cell killing and depletion-recovery kinetics. Any detectable loss of cells as a result of radiation will be a measure of the radiation damage to the hematopoietic system. Monitoring the peripheral blood cell counts for extended periods of time may show the depletion and recovery kinetics of peripheral blood cells which will be measure of the degree and duration of the radiation induced damage. Depletion-recovery kinetics also report the ability of the human hematopoietic system and the whole organism to recover after radiation induced blood cell losses. To obtain data on these endpoints, we used irradiated and control mice irradiated at NASA Space Radiation Laboratory (NSRL). Blood samples were obtained at 3 weeks before the irradiation and at 7 days, and 30 days and 75 days after the irradiation. A dose of 0.4 Gy of 350 MeV/n 28Si-ions were used for these irradiations. The depletion-recovery findings show the overall changes of the blood cell counts for each mouse. We observe a statistically significant decrease of the WBC (white blood cell) counts at day 7 |

after the irradiation. This is specifically due to the reduction in the B cells counts which are the most sensitive human blood cells and comprise the majority of circulating lymphocytes in the humanized mouse. Remarkably, there is no statistically significant difference between the WBC and the B cells and T cells pre irradiation and day 30 or day 75 after the irradiation. Two important results can be derived from this data. Firstly, irradiation of a human hematopoietic system, reconstituted in mice, is affected by 0.4 Gy of 350 MeV/n 28Si-ions. Secondly, the engrafted hematopoietic system does recover after some period of time.

Additional data was obtained by the analysis of 28Si irradiated mice done at 5-7 months after the irradiation. The data analysis shows that there is no statistically significant difference between mice irradiated with 28Si ions versus control mice when assessed for human CD45 blood cell counts in the BM, spleen, and blood. Further, there is no statistically significant difference in the weight of the mice also analyzed at this point. All these results indicate that Si irradiation does not have long-term effect on the human blood cell counts in the hematopoietic compartments and most probably on the overall physiology of the mice.

Depletion-recovery kinetics after Si irradiation have never been measured in neither human blood cells or humanized mice. As such there was a requirement for a reference irradiation to provide context for the physiological significance of the Si irradiation. For this, we studied a parallel cohort of humanized mice subject to 1Gy of 250kVp X-rays. This dose, at the selected energy, is considered a standard reference radiation having RBE (relative biological effectiveness) of 1. The results show, surprisingly, a pronounced effect of the X-ray irradiation in blood cell counts. This effect was specifically significant at day 7 after the irradiation where the WBC cell numbers were reduced 5.5 fold. Again, this effect was primarily B cell losses. The number of B cells recovered slightly at day 30 after the irradiation but decreased again at day 60. All surveyed populations (WBC, T, and B cells) exhibited statistically significant changes relative to pre-irradiated baseline for these time points. As expected, the effect of the X-rays was less pronounced on the T cells. A separate cohort of X-ray irradiated humanized mice and controls was investigated for blood cell counts between 6 and 8 months after irradiation with 1Gy of X-rays. Not surprisingly, the percent human cells in the spleen of the irradiated mice was an average of 14% while the percent of the human cells in the spleen of the control mice was an average of 47%. Interestingly, the T cell/B cell ratio was two times higher in the irradiated mice in comparison with the non-irradiated. These results show that mice can recover from X-ray irradiation, but the human cell population decreases significantly.

Overall these results indicate that for blood cell killing, the RBE of 0.4 Gy of 350 MeV/n 28Si-ions is lower than the RBE of 1Gy of 250kVp X-rays. This may be attributable, to some extent, to the relatively low particle fluence of the 28Si-ions irradiation which is around 4 particles per cell at this energy and dose. Nevertheless, this is a surprising result and shows the importance of in vivo experiments. These radiation response results are the first to show a depletion-recovery kinetics of human cells in the same humanized mice utilized in parallel for high LET exposure response studies. We believe that the obtained data will improve the space radiation risk assessment.

Assessment of early and late effects of high LET 28Si ion irradiation on the depletion and recovery kinetics of human hematopoietic stem cells in humanized mice: The array of cell types and relative population proportions of the human hematopoietic system are a function of the activity of the hematopoietic stem cells (HSC). Obviously the effect of radiation on these cells is most significant. To test the effect high-LET radiation on the HSC, we have obtained HSC counts and their relative abundance as a fraction of the lineage negative population in the BM. The cells that we focused on are characterized by the following surface markers: hCD45+, Lin-, CD34+, and CD 38-. The cells that are defined by these markers include the HSC, MPP (multipotent progenitors), and MLP (multilymphoid progenitor) cells. These cell types give rise to the lineage negative progenitors that from their side give rise of the different blood cell lineages. For simplicity here we define this group as HSC. We determined the HSC in the BM of mice irradiated with 0.4 Gy of 350 MeV/n 28Si-ions and control mice at day 7, day 30, day 60 after Si irradiation. We found statistically significant reduction of the HSC between the irradiated and control mice at all the indicated time points. An important question was when (if ever) the difference between irradiated and control mice is reduced to non-significant. To answer to this question, we accessed the HSC population in the BM of Si irradiated and control mice at 5 and 7 months after the irradiations. At this later time, no statistically significant difference between the irradiated and control mice is observed for either endpoint. In addition, there was no statistically significant reduction of human CD45 cells (WBC) in the BM of irradiated and control mice. Overall, these results show that large proportion of the BM HSC are killed by the Si ion irradiation and recover at lower pace. However, the results from the blood depletion and recovery kinetics show that the peripheral blood cells are back to the pre-irradiation counts at day 30 after the irradiation. This means that the HSC in the BM are capable of maintaining the peripheral blood cell counts even if a their counts are reduced. Taken together, these results indicate that the human hematopoietic system is capable of recovery after high-LET radiation, as already shown for low-LET radiation during HSC transplantation of tumor patients. While resilient and capable of regenerating various populations of human blood cells in the host mouse, we do observe depletion of human HSC numbers in vivo in response to high-LET radiation.

Assessment of the late effects of high LET 28Si ion irradiation on the developmental and tumor related endpoints: To test the long term effects of 0.4 Gy of 350 MeV/n 28Si-ions radiation on the human hematopoietic system, we performed very extensive blood cell population profiling of the BM of the humanized mice. There were three goals of this population analyses. The first one was to compare to what degree the irradiation changes the cell composition and the differentiation patterns of the human hematopoietic system in the mouse BM and spleen at long times after the irradiations. The second goal was to identify any aberrant cells which may indicate the initiation or the development of tumors. The third goal was to combine these results with the depletion-recovery results of the HSC and blood cells in order to build a complete picture of the effects of radiation on the human hematopoietic system. In order to conduct this profiling and attain these longer duration tumor development goals, we used flow cytometry evaluation of cells from BM and spleen from mice at 5-7 months after the irradiations.

Antibody Panel Design

Four antibody panels (two for BM and two for spleen, 8 markers in each) were used. In each panel, we selected at least one antibody for population identification (e.g., CD45). Additional antibodies selected allowed either separation of different cell lineages, or assessment of clonality, or identification of aberrant antigen expression.

Furthermore we added antibodies that are either of same lineage and same maturational stage (to identify appropriate and inappropriate expression levels), or of same lineage but different maturational stage (to ascertain normal maturation or aberrancies/ dyssynchronous expression). These panels allowed for detection and monitoring of the development of maturing myeloid elements, monocytes, blasts, immature B cells, naive B cells, and mature B cells. The survey included

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pan T-cell antigen as well as activation markers allowing a thorough evaluation of T- cells, including CD4 CD8 ratios, NK cells, the detailed evaluation of myeloid/granulocytic and monocytic maturation as well as aberrancies/discrepancies during maturation. Importantly, these panels can identify more than 20 different types of tumors.

Results

The central finding of this survey is no detectable differences between the Si-irradiated mice and the controls. The human cell counts were statistically similar in the BM, the spleen, and the blood of the irradiated and control mice. The differentiation of both lymphocyte and myeloid lineages was normal. Importantly, no aberrant marker expression was detected. These results in combination with the result from the blood and the HSC cell depletion and recovery kinetics build a very detailed picture of the effect of high LET radiation on developing human HS that we believe is first of its kind for a radiation study. We monitored the entire process of hematopoietic cell development from BM to the spleen and blood in the greatest resolution currently achievable. Combined, these results give a basis for the conclusion that the effects of high LET radiation at the studied dose and energy is not detrimental to the human hematopoietic system at least on developmental level. The damage induced initially by the irradiation did not persist in our system of study. Also the lack of tumor development shows that at this energy and dose and time point, high LET radiation is unable to transform human cells in vivo.

Cytogenetic evaluation of the chromosomal damage induced by high and low LET radiation: Dicentric chromosome formation.

The goal of the Dicentric chromosome formation analysis was to estimate the chromosomal damage in relation to the dose, energy and the RBE of the radiation types used. Such data would allow us to directly compare the radiation induced damage. In addition, we characterized the damage in a timely manner. The goal here was to monitor persistence of radiation induced chromosome damage. A dose of 0.4 Gy of 350 MeV/n 28Si-ions and a dose of 1Gy X-rays was used for these irradiations. Dicentric formation was accesses at day 7, day 30, and day 60 after the irradiations in human T cells purified from blood and spleen of humanized mice. In preliminary experiments we have shown that there is no difference in the radiation induced dicentric chromosome formation in cells purified from blood or spleen of humanized mice. Three mice per time point were used. At least 500 metaphases were analyzed for each time/dose point.

The results show that 28Si irradiation is twice less effective in dicentrics chromosome induction. This is somewhat surprising result since we expected that high–LET particle radiation should be more effective in DNA damage induction than photons. This data, although surprising, harmonizes with the data for blood cell killing obtained in this study, which shows that the X-rays are more effective.

The second observation here is that the numbers of dicentrics decreased in time dependent manner for both radiation types. This is not an unexpected result given prior work showing dicentric frequency reduction over time in the blood of irradiated individuals.

Two papers with the results from the study are currently in final stages of preparation.

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Our main hypothesis is that studies of the effects of high-LET radiation on a human hematopoietic system, reconstituted in immunodeficient mice, will provide meaningful information on not only post-radiation depletion and repopulation kinetics but also tumor formation. Such information can then be applied to space radiation risk assessment. The importance of this work is clear given the lack of in vivo data on the effects of high LET radiation on human organ and tissue systems. The study has two key objectives: The first is to characterize the early and late response of a human immune system reconstituted in mice to high LET radiation and gamma-rays. The goal here is to build a detailed picture of the series of changes that the reconstituted human hematopoietic system undergoes after a high LET radiation exposure. The second objective is to characterize the human tumors that may arise in the irradiated and non-irradiated mice. To accomplish these objectives we produced and irradiated mice engrafted with a human hematopoietic system. The mice were exposed to either HZE (high energy particles) ions or X-rays and subsequently monitored along with non-irradiated control mice. The early effects of the radiation on the human cells are assessed by periodic peripheral blood tests. The obtained differential counts are compared with differentials acquired before the irradiation. Monitoring the peripheral blood cell counts for extended periods of time may show the depletion and recovery kinetics of peripheral blood cells which will be a measure of the degree and duration of the radiation induced damage and also the ability of the human hematopoietic system to recover after radiation induced insults. In addition, the mice are monitored for tumor development. The tumor characterization was done by a team of pathologists from the Columbia University Department of Pathology and Cell Biology using established clinical laboratory methods for human tumor characterization with corrections based on the specificity of the mouse model. This project directly addresses high-priority research topics emphasized by the Space Radiation Program Element for FY2013 and in in particular the Risk of Radiation Carcinogenesis from Space Radiation and Risk of Degenerative Tissue or other Health Effects from Space Radiation.

The following endpoints were studied in the reported period:

1. Early effect studies: Detailed analysis of the short term effects of high LET 28Si ions irradiation on depletion and recovery kinetics of human hematopoietic stem cells in mouse bone marrow

2. Early effect studies: Determination of the in vivo induction of dicentric chromosomes in human cells from mouse blood by high-LET radiation

3. Late effect studies: Complex analysis of high-LET irradiated mice for developmental tumor related endpoints

Early effect studies: Depletion and recovery kinetics of human hematopoietic stem cells (HSC) in the mouse BM after irradiation with Si ions. With these experiments we were aiming to estimate how the engrafted in the mouse bone marrow human hematopoietic stem cells counts will be affected shortly after irradiation with high LET irradiation. The importance of these studies comes from the fact that the hematopoietic stem cells spawn the array of cells that comprise the immune system. Any changes in their counts and functions affect the whole system as such.

We performed a detailed immunophenotyping analysis of human hematopoietic stem and progenitor cells. The cell counts were obtained at three time points – 7 weeks, 15 weeks, and 22 weeks after the irradiation. Five mice with engrafted cells from the same human donor (3 irradiated and 2 controls) per time point were used. The results showed initial reduction of the human hematopoietic stem cell counts by factor of 2, followed by slow recovery. The HSC

counts in the irradiated mice normalized to levels seen in control mice around 20-22 weeks after the irradiation. In contrast, the peripheral blood cell counts returned to their normal values in about 30-45 days. This important comparison shows that even lower than normal numbers of human hematopoietic stem cells, can still support and replenish normal peripheral blood cell populations. These results are first to show in vivo that the human hematopoietic system is affected by 0.4 Gy of 350 MeV/n 28Si-ions Si irradiation and that it can recover in a relatively short period of time after being exposed. These results may be used for risk estimation of astronauts in deep spaceflights.

Early effects: Dicentrics analysis. In addition to cell counts, an important methodology used frequently in radiation biodosimetry is the estimation of dicentric chromosome frequency as a results of radiation. In contrast to the cell counts it reports a specific type of chromosomal damage. Since chromosome dicentric frequency correlate very well with the radiation dose, the results can be used to compare the effects of different types of radiation. In our studies we measured the frequency of dicentric chromosomes in human T cells isolated from humanized mice 7 days after irradiation with 0.4 Gy of 350 MeV/n 28Si ions. The results of the analysis of more than 500 metaphases show dicentric frequencies of 0.06. These results are the first to show the induction of dicentric chromosomes in vivo by 28Si-ion irradiation and can be used in comparative studies for relative biological effectiveness of different types of radiation.

Late effects: Developmental and tumor related endpoints. A detailed analysis of the human hematopoietic system reconstituted in irradiated and control, was done between 7 and 10 months after the irradiation. This included morphological and cytogenetic analysis aimed to show the effect of radiation on tissue and chromosomal level, a detailed developmental analysis aimed to detect radiation induced abnormalities in human cell development done by immunophenotyping and a detailed tumor related analysis aimed to detect tumor development in the mice also done by immunophenotyping.

The results from the morphological analysis done on femurs from irradiated and control mice show no differences in the bone marrow morphology between irradiated and control mice. For both groups of mice, the mouse bone marrow was healthy and active in cell production. The cytogenetic analysis done by conventional (G-band) karyotyping identified diploid male and female karyotypes without abnormalities. Most extensive was the immunophenotypic analysis done on human cells from mouse bone marrow and spleen. We employed panels comprising 20 different markers specific for developmental and tumor related processes. The combination of markers in the different panels allowed for the detection of more than 20 different hematological tumors. Each sample was stained with 4 panels of 8 markers and processed by flow cytometry. The results show, with high resolution, different stages in human cell development starting from HSC and continuing through the early and late progenitor stages toward terminally differentiated cells. The comparison between samples collected from irradiated and non-irradiated mice show no developmental abnormalities between the two groups. A careful analysis for tumor development show no immunophenotypic features diagnostic for a lymphoproliferative or myeloid disorder or neoplasm. Since tumor development usually takes long time, a sorted bone marrow human CD34+ cells from irradiated and control mice were engrafted in new mouse hosts for further monitoring. These mice are currently being monitored for tumor development.

| **Bibliography Type:** | Description: (Last Updated: 08/25/2020) |
|---|---|
| **Articles in Peer-reviewed Journals** | Hoehn D, Pujol-Canadell M, Young EF, Serban G, Shuryak I, Maerki J, Xu Z, Chowdhury M, Luna AM, Vlad G, Smilenov LB. "Effects of high- and low-LET radiation on human hematopoietic system reconstituted in immunodeficient mice." Radiat Res. 2019 Feb;191(2):162-75. https://doi.org/10.1667/RR15148.1 ; PMID: 30520704; PMCID: PMC6555159 , Feb-2019 |
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